ONTARIO TEJ4M COMPUTER ENGINEERING TECHNOLOGY

HARDWARE REQUIREMENTS SPECIFICATION

PROJECT

Bland2Grand

Smart Autonomous Carousel Spice Dispensing Appliance

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1 Executive Summary

1.1 Project Overview

Bland2Grand is a smart, autonomous countertop spice dispensing appliance that eliminates the guesswork of seasoning food. A user types the name of any dish into a web application on their phone, the system suggests appropriate spice blend options drawn from a local recipe database, the user selects their preferred blend and serving size, and the machine autonomously dispenses the precise gram-accurate amount of each required spice directly into a bowl, with no manual measurement required.

The machine communicates wirelessly over WiFi, operates from a single 24V DC power supply stepped down to 5V where required, and needs no technical knowledge to use. The physical product is a 3D-printed tower enclosing a motorised octagonal carousel. Eight interchangeable spice modules are seated one on each face of the octagon. A NEMA 23 stepper motor, monitored by an AS5600 magnetic encoder, rotates the carousel platform to index the correct spice module under a fixed output chute. Once the target module is in position, a NEMA 17 stepper motor descends onto the module's exposed auger shaft and drives the internal auger screw, metering spice through the chute into a bowl below. A NOYITO 1 kg load cell with HX711 24-bit ADC provides real-time gram feedback to a closed-loop control algorithm running on an Arduino Uno R4 WiFi. The Uno R4 WiFi communicates with a Flask web server hosted on a laptop over the onboard 2.4 GHz WiFi module built into the RA4M1 microcontroller. No external ESP8266 module is required.

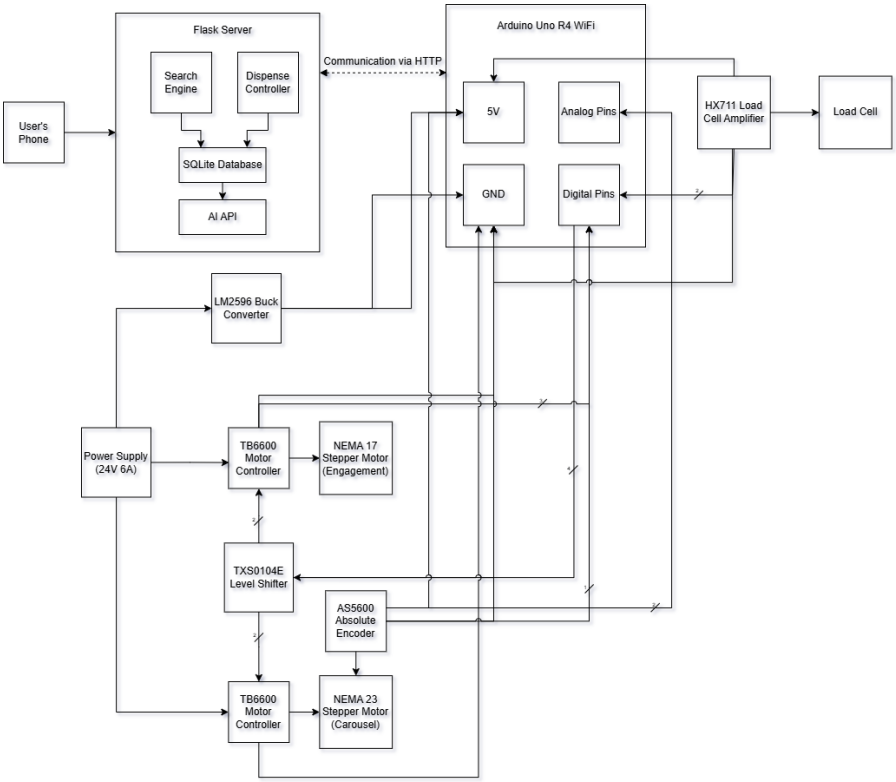

Figure 1: Bland2Grand High-Level System Block Diagram

The target audience spans two distinct market segments: Beginner home cooks who lack the culinary experience to season food correctly; and small commercial or food-service operations, such as catering businesses, meal-prep kitchens, and restaurant prep stations, where consistent and repeatable spice blending at scale is operationally valuable. The technical architecture supports both use cases from the same hardware, with single-serving dispenses for the home user and large-batch scaled dispenses for the commercial user.

1.2 Purpose and Scope of this Specification

This Hardware Requirements Specification (HRS) defines all functional, mechanical, electrical, software, and interface requirements for the Bland2Grand system. It serves as the authoritative design contract between the student designer (Elliott Starosta) and the course teacher (Mr. Roller), who acts in the capacity of project manager and technical reviewer. The client is a notional end-user representing the two market segments described above. The end user is assumed to be a person with no engineering or electronics background who interacts with the system exclusively through a phone-based web interface.

The scope of this specification covers the complete hardware and software system from power inlet to spice outlet. It includes all mechanical assemblies, all electronic circuits, the Arduino firmware, the Flask backend, and the mobile web interface. It does not cover industrial certification, food-safety regulatory compliance, or long-term reliability testing.

All curriculum expectations of the Ontario TEJ4M Computer Engineering Technology course must be demonstrably satisfied by this project. The relevant curriculum strands addressed are A2 (Computer Systems), A3 (Electronics, Robotics, and Computer Interfacing), A4 (Networking Concepts), A5 (Data Representation and Digital Logic), B3 (Electronics Build), B4 (Network Skills), and B5 (Programming). Specific curriculum alignment is noted throughout Section 3 where applicable.

The project build timeline runs from April 1, 2026 to the final submission and demonstration on June 7, 2026, a period of approximately ten weeks. Phase 1 (Weeks 1–2) covers electronics breadboarding: carousel motor, encoder, and engagement motor integration with load cell. Phase 2 (Weeks 3–4) covers Flask backend and WiFi communication. Phase 3 (Weeks 5–7) covers mechanical design, 3D printing, and module assembly. Phase 4 (Weeks 8–9) covers full system integration and closed-loop tuning. Phase 5 (Week 10) covers testing, calibration, documentation, and demonstration preparation.

2 Product Description

2.1 Product Context

Bland2Grand is a standalone, self-contained appliance. It does not depend on any cloud infrastructure during normal operation. The recipe database is stored locally in SQLite on the host laptop, and all dispensing logic runs on the Arduino. The only external dependency during operation is a local WiFi network that allows the phone and the laptop to communicate over HTTP. An optional AI API fallback (OpenAI or Anthropic) is called over the internet only when a user requests a dish not present in the local database; this

fallback also degrades to a dish-not-found message if internet connectivity is unavailable.

The product occupies a gap in the market that no known consumer product currently addresses. Professional-grade automated dispensing systems used in commercial food production exist but cost tens of thousands of dollars and are inaccessible to individual cooks or small businesses. Manual spice racks require the user to know exactly what to measure and how much. Bland2Grand sits between these two extremes: an intelligent single-axis carousel with gram-accurate closed-loop dispensing at a sub-$200 hardware cost.

2.2 User Characteristics

Two user profiles have been developed for this project.

User Profile A: The Beginner Home Cook. This user is typically a student, young professional, or first-time cook. They can cook basic proteins and starches but consistently produce under-seasoned or inconsistently flavoured food because they do not know how much of each spice to use or how spice flavours interact. They are comfortable with a smartphone web app and expect interactions to feel like consumer software rather than a technical tool. They have no electronics or programming knowledge and should never be required to interact with the hardware beyond placing a bowl and tapping their phone.

User Profile B: The Small Commercial Operator. This user runs a catering company, meal-prep kitchen, or small restaurant prep station. They prepare large batches of seasoned food and need consistent, repeatable spice ratios across every batch. Inconsistent seasoning is a business risk. This user may want to define and store custom house recipes in the system. They value speed and accuracy over novelty and are primarily interested in the machine as a productivity and quality-control tool. The serving-size scaling feature is critical to their workflow.

2.3 Assumptions and Risks

Assumption 1: A local WiFi network (2.4 GHz 802.11 b/g/n) is available at the demonstration site. The Arduino Uno R4 WiFi module does not support 5 GHz. If only 5 GHz is available, a mobile hotspot on 2.4 GHz will be used as contingency.

Assumption 2: A 3D printer capable of printing PLA with a minimum 0.2 mm layer height and 0.4 mm nozzle diameter is available at school. The auger mechanism and carousel indexing features require tolerances of 0.2 mm, achievable on most FDM printers but likely requiring multiple calibration prints.

Assumption 3: The NEMA 17 (17HS4401S) and NEMA 23 (23HS5628) stepper motors, TB6600 driver, AS5600 magnetic encoder, and DC-DC buck converter modules can be sourced and delivered within Canada before April 1, 2026.

Assumption 4: The host laptop running the Flask backend will remain powered and connected to the local WiFi network for the entire duration of any dispense session. If the laptop sleeps or disconnects mid-session, the Arduino firmware watchdog timeout (HRS-3.1.44) will halt all motors within 30 seconds.

Assumption 5: All 8 spice channels will be loaded with the specific spices defined in Section 3.1.8 during calibration and demonstration. Swapping a spice without re-running the per-channel calibration will result in inaccurate gram dispenses due to bulk density differences.

Assumption 6: The demonstration environment will have a stable, level surface for the tower. The load cell platform must remain level to within 2 degrees throughout a dispense session.

Assumption 7: The AI API provider (OpenAI or Anthropic) will remain available and responsive for the duration of the demonstration. Because the AI fallback is non-critical, this assumption carries low project risk; the Flask backend shall display a graceful error message if the API returns an error or times out.

Assumption 8: The AS5600 magnetic encoder and its diametrically magnetised disc magnet will align to within the 0.5 mm axial positioning tolerance required for reliable angle readout. A 3D-printed magnet holder will be designed to maintain this alignment on the carousel shaft.

Risk 1: The NEMA 23 carousel motor may produce vibration that introduces noise in the HX711 load cell reading. *Mitigation:* The carousel only rotates during positioning, never during active dispensing. The firmware enforces a mandatory 1-second settling delay after the carousel stops before initiating any dispense.

Risk 2: The NEMA 17 engagement motor coupling to each module's exposed auger shaft requires a precise mechanical interface. If the coupling slips or misaligns, the auger will not turn or will turn inconsistently. *Mitigation:* A D-shaft profile or hex-socket coupler is designed into the top of each module's auger shaft, and a compliant spring-loaded coupler on the NEMA 17 output shaft compensates for minor vertical alignment errors.

Risk 3: Fine spice powders (e.g., garlic powder, bulk density approximately 0.49 g/mL) may compact inside the auger tube over time. *Mitigation:* The auger tube diameter and helix pitch are sized generously for the finest expected powder, and per-spice calibration accounts for density differences.

Risk 4: The AS5600 encoder operates on 3.3 V I2C logic. The Uno R4 WiFi also operates at 3.3 V I/O, so no level shifting is required. However, if the LM2596 buck module output drifts, it could over-voltage the encoder. *Mitigation:* The LM2596 output is verified and trimmed to 5.0 V $\pm$0.1 V before connection to the distribution rail, and the 3.3 V for the encoder is sourced from the Uno R4 WiFi onboard 3.3 V regulator.

Risk 5: The designer is not yet fully proficient in CAD software. *Mitigation:* The designer will complete sufficient training to reach a proficient standard prior to April 1, 2026.

Risk 6: Driving the NEMA 23 at 2.8 A through the TB6600 at 24 V produces significant heat in the driver. *Mitigation:* The TB6600 has an onboard heatsink and thermal protection. Current limit on the driver will be set conservatively to 2.0 A since the carousel positioning load is light and does not require full rated torque.

Risk 7: The carousel octagon platform introduces a longer total dispense cycle compared

to a parallel 8-motor design because the carousel must index between spices sequentially. *Mitigation:* Pre-analysis shows that at an indexing speed of 60 RPM, the worst-case carousel rotation between two adjacent modules (45 degrees) takes approximately 0.75 seconds, which is acceptable within the 120-second total dispense time requirement.

2.4 Constraints

Budget Constraint: Total component cost must remain at or below $200 CAD. The current bill of materials is estimated at approximately $165 CAD.

Microcontroller Constraint: The system uses a single Arduino Uno R4 WiFi board. All motor control, encoder reading, scale reading, and WiFi communication must be handled by this one board.

Pin Count Constraint: The Arduino Uno R4 WiFi provides a limited number of I/O pins, including 14 digital I/O pins and 6 analogue input pins. Depending on the amount of hardware integrated into the system, this limited pin count may become a design constraint. If additional peripherals require more I/O connections than are available, an external pin expansion device (e.g., an I^2C GPIO expander such as the MCP23017) may be required to increase the number of usable pins.

Software Language Constraint: The Arduino firmware must be written in C++ using the Arduino IDE or PlatformIO. The Flask backend must be written in Python 3. The web interface is HTML, CSS, and JavaScript served by Flask.

Mechanical Constraint: All structural components are 3D printed in PLA using school equipment. No CNC machining, laser cutting, or custom PCB fabrication is within scope.

Curriculum Constraint: The project must satisfy the TEJ4M curriculum expectations listed in Section 1.2.

Sequential Dispensing Constraint: The firmware shall dispense spices one channel at a time. The carousel-and-single-auger-motor architecture enforces this inherently: Only the module currently indexed under the engagement motor can dispense. This also simplifies the closed-loop weight control, as the HX711 always measures weight change from a single active auger.

Network Topology Constraint: The Uno R4 WiFi must operate in WiFi station mode on the same local subnet as the host laptop. Access point (AP) mode is not permitted as it would require the user's phone to disconnect from the school or home router and break AI API internet access.

Power Architecture Constraint: A single 24 V DC supply powers the NEMA 23 carousel motor and the NEMA 17 engagement motor through two seperate TB6600 drivers. All 5 V logic components (HX711, Uno R4 WiFi) are powered via the LM2596 DC-DC buck converter stepped down from 24 V to 5 V.

Filament Colour Constraint: All structural and cosmetic tower components shall be printed in a single consistent filament colour. Functional internal components (auger screws, shaft couplers) may be any available colour as they are not externally visible.

Open-Source Constraint: All software components must be original code written by the student designer (Elliott Starosta), with the exception of imported open-source libraries. This satisfies the B5 (Programming) curriculum strand.

2.5 Dependencies

Dependency 1: Encoder Validation Before Carousel Integration. The AS5600 encoder readout must be verified on the bench (correct angle values reported over I2C) before the carousel motor assembly is mechanically integrated into the tower.

Dependency 2: Engagement Motor Coupling Validation Before Module Integration. The NEMA 17 engagement motor and shaft coupler must be confirmed to reliably engage and spin the auger shaft on a single test module before all 8 modules are assembled.

Dependency 3: Flask Serial Communication Before WiFi Development. Arduino dispensing firmware and the Flask backend must first be validated communicating over USB serial before transitioning to WiFi.

Dependency 4: Load Cell Calibration Before Recipe Database Population. Perspice calibration values must be measured empirically before the recipe database can be populated with accurate gram amounts.

Dependency 5: 3D Print Iteration Before Final Assembly. At least one full test print of each unique mechanical part must be completed and dimensionally verified before production prints begin. Tolerance adjustments are expected particularly for the carousel indexing feature, the auger shaft coupler socket, and the engagement motor mount.

3 Requirements

3.1 Functional and Hardware Requirements

Requirements are numbered with the prefix **HRS-3.1.x**. The **Pri.** column records the priority level: **H** (High) for requirements essential to a functional demonstration, **M** (Medium) for requirements strongly desired, and **L** (Low) for stretch goals if time permits.

Table Column Guide

Column	Description
Req#	Unique requirement identifier.
Requirement	Normative *shall* statement describing the system behaviour.
Comments	Rationale, analysis, assumptions, or design notes related to the requirement.
Pri.	Priority level: **H** (High), **M** (Medium), **L** (Low).
Date Rvwd	Date the teacher reviewed the requirement.
Customer Rev. / Apvd	Teacher sign-off initials indicating approval.

3.1.1 Overall System

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.1	The system shall autonomously dispense a user-specified spice blend into a bowl placed on the integrated scale platform, requiring no manual measurement by the user.	Core value proposition. Hands-free dispensing for all user profiles. The carousel indexes to each required spice module in sequence; the engagement motor drives that module's auger.	H		
HRS-3.1.2	The system shall support a minimum of 8 independent spice channels, one per face of the octagonal carousel platform.	8 modules selected to maximise achievable blends. The carousel format replaces 8 individual drive motors with a single carousel motor and a single shared engagement motor.	H		
HRS-3.1.3	Each dispense operation shall achieve a target weight accuracy of ± 0.3 g for any individual spice amount between 0.5 g and 50 g.	Verified empirically during calibration testing with at least 10 consecutive dispenses per spice channel.	H		
HRS-3.1.4	The system shall automatically scale all spice amounts proportionally when the user specifies a serving count between 1 and 20.	Scaling is performed in the Flask backend prior to issuing dispense commands to the Arduino.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.5	The total time to complete a full 8-spice blend dispense for a single serving shall not exceed 120 seconds under normal operating conditions.	Measured from user pressing Dispense to the final confirmation message on the phone. 120 s accounts for carousel indexing time between modules (≈0.75 s per transition) plus per-spice dispense time.	M		
HRS-3.1.6	The system shall operate from a single 24 V DC 6 A power supply. The NEMA 23 motor, NEMA 17 motor, and both TB6600 drivers shall run directly from 24 V. All 5 V logic components shall be powered via an LM2596 DC-DC buck converter stepped down to 5 V.	Centralised 24 V architecture allows the NEMA 23 and NEMA 17 to operate at its rated voltage. The LM2596 provides up to 3 A at 5 V, sufficient for the HX711 and Uno R4 WiFi (500 mA). The AS5600 encoder is powered from the Uno R4 WiFi's onboard 3.3 V regulator, not from the LM2596 output directly.	H		

3.1.2 Mechanical Chassis and Carousel

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.7	The tower enclosure and all carousel components shall be entirely 3D-printed in PLA filament using school FDM printing equipment.	PLA chosen for dimensional stability and ease of printing on school-grade hardware. Operating temperature is room temperature, well within PLA's heat deflection point of ≈60 ℃.	H		
HRS-3.1.8	The tower enclosure shall have overall dimensions not exceeding 300 mm in width, 300 mm in depth, and 400 mm in height.	Must fit on a standard school desk and a typical kitchen counter. The carousel architecture is wider than a single-tower design and the footprint constraint reflects this.	H		
HRS-3.1.9	The carousel base shall be an octagonal rotating platform driven by the NEMA 23 stepper motor. Each of the 8 faces of the octagon shall accept one spice module.	Octagonal geometry provides 45-degree index angles between adjacent modules, minimising carousel rotation distance for any two-module sequence.	H		
HRS-3.1.10	Each spice module shall be mechanically interchangeable with any of the 8 carousel faces. All 8 modules shall be identical in external geometry and auger shaft interface.	Enables easy module replacement. Only one module CAD file needs to be designed and printed 8 times.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.11	Each module shall lock onto its carousel face via a quarter-turn bayonet or clip mechanism, enabling tool-free removal and insertion while the carousel is stationary.	Bayonet chosen over snap-fit because snap-fits fatigue after repeated cycling. The bayonet distributes rotational force across the full circumference.	H		
HRS-3.1.12	The auger screw inside each module shall be printed with a radial clearance of 0.2 mm between the outer diameter of the auger helix and the inner diameter of the auger tube.	Prevents passive leakage of fine powder while keeping friction low enough for the NEMA 17 to drive without stalling. Tighter than 0.1 mm risks binding; looser than 0.3 mm risks passive leakage.	H		
HRS-3.1.13	The auger housing tube inside each module shall be oriented at 20 degrees above horizontal from the input end to the nozzle end.	Upward angle means spice must travel uphill to exit. Gravity acts as a passive valve when the engagement motor is stopped, preventing free-flow dripping. 20° is the midpoint of the empirically recommended 15–25° range for food-powder auger dispensers.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.14	Each module's auger shaft shall protrude from the top face of the module as a D-shaft or hex-profile stub of minimum 8 mm engagement length. This stub shall accept the spring-loaded coupler on the NEMA 17 engagement motor output shaft.	The standardised shaft stub profile ensures every module interfaces identically with the engagement motor. The 8 mm engagement length ensures torque transfer without slipping even at the NEMA 17's maximum rated torque. Note: because the auger tube is angled at 20° (HRS-3.1.13), the auger shaft inside the module is also angled. The exposed stub at the top of the module shall be designed with a bevel or universal joint interface so that it presents a vertically-oriented engagement face to the NEMA 17 output shaft, which is mounted vertically downward per HRS-3.1.26. This geometry shall be verified in CAD before printing.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.15	The auger shaft shall include a circular flange disc at the motor-end of the helix section, fitting the auger tube with 0.1 mm radial clearance.	Acts as a physical seal preventing spice from migrating toward the top shaft stub. Without this seal, fine powder packs around the shaft and clumps, causing binding within hours of use.	H		
HRS-3.1.16	The spice chamber above each auger shall include a vertical window slot or transparent wall section no less than 40 mm tall for visual inspection of remaining spice level without opening the module.	Allows the user to assess refill needs at a glance without stopping the machine.	L		
HRS-3.1.17	The fixed tower frame shall include a single output chute aligned with the engagement motor position. All modules, when indexed to the active position, shall align their nozzle with this chute. The chute shall direct spice into the centre of the scale platform below. Chute walls shall be angled at no less than 45 degrees from vertical.	45° is the minimum angle for free powder flow. Steeper angles (60–70°) preferred for fine powders such as garlic powder.	H		
HRS-3.1.18	All 3D-printed parts in the direct spice flow path shall be printed at no less than 50% infill density to ensure non-porous walls.	Porous walls allow hygroscopic spices to absorb ambient moisture through the wall material and clump over time.	M		

3.1.3 Carousel Motor and Encoder

The NEMA 23 stepper motor with TB6600 driver was selected to rotate the carousel platform. NEMA 17 motors were evaluated and rejected for the carousel drive: The carousel platform carries the mass of 8 loaded spice modules (estimated 800 g–1.2 kg total) on a bearing, and a NEMA 17 at 12V may not provide sufficient holding torque to index repeatably without step loss. The NEMA 23 at 24V delivers significantly higher torque, ensuring reliable indexing regardless of module load.

The AS5600 12-bit magnetic rotary encoder is used to provide absolute angular position feedback of the carousel shaft. This enables the firmware to confirm that the carousel has reached the correct index angle for each module, correcting any step loss and allowing the system to home itself on startup to a known reference position.

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.19	The carousel shall be driven by a NEMA 23 (23HS5628) stepper motor rated at 2.8 A, paired with a TB6600 microstepping driver. The TB6600 current limit shall be set to 2.0 A in operation to reduce heat dissipation.	NEMA 23 provides sufficient torque margin to rotate a fully loaded carousel platform (up to 1.2 kg) without step loss. The TB6600 supports up to 4 A and accepts 9–42 V input, fully compatible with the 24 V supply.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.20	The TB6600 shall be connected to the Arduino Uno R4 WiFi on 2 digital output pins: STEP on pin 3 (PWM-capable) and DIR on pin 4. The TB6600 ENA pin shall be tied high (enabled) by default, keeping the carousel motor continuously energised and holding position between moves.	2-wire step/direction interface is standard for TB6600. Keeping ENA permanently high means the carousel motor coils are always energised, providing continuous holding torque to prevent the loaded carousel from drifting under gravity between index moves. This is a deliberate design decision: the carousel load (up to 1.2 kg) and potential vibration from the engagement motor make continuous hold preferable to power-saving de-energisation. The resulting steady-state thermal load on the carousel TB6600 is acceptable given the 2.0 A current limit setting and the driver's onboard heatsink.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.21	The AS5600 magnetic encoder shall be mounted coaxially on the carousel motor shaft, with the diametrically magnetised disc magnet affixed to the shaft end within 0.5 mm axial air gap. The encoder shall communicate with the Uno R4 WiFi via I2C (SDA on pin A4, SCL on pin A5) at 400 kHz.	The AS5600 provides a 12-bit angle value (0–4095 over 360°), yielding 0.088° per count. The 45° inter-module index step corresponds to 512 encoder counts, giving substantial noise margin.	H		
HRS-3.1.22	The firmware shall use the AS5600 encoder to verify carousel position at the end of every indexing move. If the measured angle deviates from the target by more than ±1.0°, the firmware shall issue corrective steps until the position is within tolerance before engaging the auger drive motor.	Closed-loop carousel positioning prevents spice from being dispensed through the wrong module if mechanical slip or inertia overshoot causes the carousel to stop short of the target index angle.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.23	On power-up, the firmware shall perform a carousel homing routine by slowly rotating until the AS5600 encoder reports the absolute angle corresponding to Module 1. Module 1 shall be designated as the zero reference for all subsequent indexing operations.	Absolute encoder homing eliminates the need for a physical limit switch. Because the AS5600 reports absolute angle (not incremental pulses), the system always knows its position, even after a power cycle.	H		
HRS-3.1.24	The firmware shall enforce a minimum 1-second settling delay after the carousel motor stops before enabling the NEMA 17 engagement motor and initiating a dispense.	Ensures all carousel vibration has dampened and the load cell reading has stabilised before dispensing begins. Prevents inertial overshoot of the carousel from affecting the module-to-chute alignment during dispensing.	H		

3.1.4 Engagement Motor and Auger Drive

The NEMA 17 (17HS4401S) stepper motor was selected as the engagement motor that couples to each module's auger shaft. This motor descends onto the exposed auger stub from above the carousel, driven by the module's internal auger helix, metering spice through the fixed output chute.

NEMA 17 was chosen over a DC motor for the same reason as in all auger-dispensing contexts: discrete step control provides a firm, instantaneous stop with no coasting, minimising overshoot. NEMA 17 was chosen over NEMA 23 for the engagement motor because the auger load is light, the motor only needs to overcome the friction of dry spice powder inside the auger tube, and NEMA 17's smaller form factor simplifies the engagement mechanism mount above the carousel.

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.25	The engagement motor shall be a NEMA 17 (17HS4401S) stepper motor, 4-lead, 1.5 A, paired with a TB6600 microstepping driver on the 24 V rail.	17HS4401S: 1.5 A/phase, 1.8° step angle, 45 N·cm holding torque. This torque is 10× in excess of the auger driving requirement, providing substantial margin for variable powder compaction. The TB6600 runs on the 24 V rail, consistent with the carousel motor driver.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.26	The NEMA 17 engagement motor shall be mounted on a fixed vertical rail above the carousel, with its output shaft oriented vertically downward so that it can directly engage the upward-facing auger stub on whichever module is indexed to the active position.	Fixed mounting eliminates any need for a vertical actuation mechanism. Because the auger shaft stub is at the top of each module and all modules are identical in height, the vertical distance between the NEMA 17 output shaft and the auger stub is constant for all 8 modules. The spring-loaded jaw coupler (HRS-3.1.27) provides the engagement and disengagement mechanism: when the carousel rotates to a new module, the lateral motion of the stub passing under the coupler jaw causes the coupler to automatically release and re-engage without any vertical movement of the NEMA 17 motor itself.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.27	The NEMA 17 output shaft shall connect to the module auger stub via a spring-loaded jaw coupler or flexible disc coupler with a minimum 3 mm of axial compliance. The coupler shall engage the D-shaft or hex-profile stub on the module with no tools required.	3 mm axial compliance allows for minor vertical misalignment between the motor shaft and module stub due to print tolerances or uneven seating. The spring pre-load ensures positive torque transmission without backlash. The jaw geometry shall be designed so that the coupler self-disengages cleanly when a non-active module's stub passes laterally through the engagement zone during carousel rotation, and self-engages when the next target module's stub arrives at the active position.	H		
HRS-3.1.28	The TB6600 shall be connected to the Arduino Uno R4 WiFi on 2 digital output pins: STEP on pin 5 (PWM-capable) and DIR on pin 7. The TB6600 ENA pin shall be tied high (enabled) by default.	2-wire step/direction interface is standard for TB6600. Using a PWM-capable pin for STEP allows hardware-timer-based step pulse generation, ensuring precise timing independent of software loop speed.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.29	The firmware shall implement a three-stage speed ramp-down as dispensed weight approaches the target: at 80% of target, engagement motor speed reduces to 50% of maximum; at 95%, to 15%; at 100%, the motor stops immediately and coil current is removed after 500 ms.	Graduated deceleration exploits the stepper motor's ability to hold position precisely at stop. Without this ramp, a motor running at full speed would overshoot the target due to in-flight spice remaining in the auger tube. Coil current removal after 500 ms prevents unnecessary heating of the NEMA 17 during idle periods.	H		

3.1.5 Load Cell and Weight Sensing

The NOYITO 1 kg load cell (full-bridge strain gauge) with HX711 24-bit ADC amplifier was selected for weight sensing. The rationale is identical to the original design: Volume-based dispensing is unreliable across different spice bulk densities, whereas a gram-accurate scale eliminates this variability entirely.

Resolution calculation:

$$2^{24} = 16{,}777{,}216 \text{ counts} \qquad \frac{1000\,\text{g}}{16{,}777{,}216} \approx 0.06\,\text{mg per count}$$

In practice, electronic noise floors the usable resolution to approximately 0.1 g, which is well within the $\pm$0.3 g accuracy requirement.

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.30	The system shall use a NOYITO 1 kg straight-bar full-bridge strain gauge load cell paired with an HX711 24-bit ADC amplifier.	Full-bridge configuration provides temperature self-compensation and common-mode noise rejection. The straight-bar geometry mounts cleanly under the scale platform with two M4 bolts per the datasheet mounting pattern.	H		
HRS-3.1.31	The HX711 shall communicate with the Uno R4 WiFi via its 2-wire serial protocol on pin 9 (DT/data) and pin 10 (SCK/clock).	HX711 DT and SCK are bit-banged GPIO. No hardware UART, SPI, or I2C peripheral is required beyond those already used by the AS5600 encoder.	H		
HRS-3.1.32	The firmware shall perform a tare (zero reset) operation on the load cell before each individual spice dispense operation, after the carousel has finished indexing and the 1-second settling delay has elapsed.	Taring between spices eliminates cumulative drift and accounts for settling of previously dispensed spice. The firmware measures only the weight increase attributable to the current spice.	H		
HRS-3.1.33	The firmware shall implement a 500 ms settling delay after the engagement motor stops before taking the final confirmatory weight reading.	Strain gauge load cells exhibit a brief oscillation immediately after a load change. 500 ms is sufficient for the load cell and mounting structure to stabilise.	H		

3.1.6 Power System

Power budget analysis.

Component	Voltage	Current (ea.)	Current (total)
NEMA 23 via TB6600 (set to 2.0 A)	24 V	2000 mA	2000 mA
LM2596 buck converter (quiescent + output load)	24 V in	—	200 mA (24 V side)
NEMA 17 via TB6600 (set to 1.5 A)	24 V	1500 mA	1500 mA
Uno R4 WiFi (MCU + WiFi active)	5 V	500 mA	500 mA
HX711 amplifier	5 V	2 mA	2 mA
AS5600 encoder	3.3 V	10 mA	10 mA
Total at 24 V (worst case)	**24 V**	—	**≈3.7 A**
Supply rated capacity	**24 V**	—	**6 A**

The 24 V 6 A supply provides 62% headroom over the worst-case draw of 3.7 A, ensuring stable voltage under all operating conditions.

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.34	The entire system shall be powered by a single 24 V 6 A AC/DC power supply with a 5.5 × 2.1 mm barrel jack output (100–240 V AC input).	Universal input range makes the system compatible with North American and international mains. 6 A rated capacity at 3.7 A worst-case draw provides 62% headroom, ensuring the supply runs well within its thermal envelope.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.35	The 24 V rail shall connect directly to both TB6600 drivers' power input terminals. The carousel TB6600 current limit potentiometer shall be set to 2.0 A; the engagement TB6600 current limit potentiometer shall be set to 1.5 A.	Operating the NEMA 23 at 2.0 A provides ample carousel torque and the NEMA 17 at 1.5 A matches its rated phase current, both keeping TB6600 temperatures well below the thermal shutdown threshold.	H		
HRS-3.1.36	An LM2596 DC-DC buck converter module shall step the 24 V rail down to 5 V $\pm$0.1 V to supply the Arduino Uno R4 WiFi and the HX711. The LM2596 output shall be verified with a multimeter and trimmed to 5.0 V before system assembly.	LM2596 rated at 3 A output. The 5 V load (Uno at 0.5 A + HX711 at 2 mA) totals $\approx$0.502 A, within the 3 A rating with 83.3% headroom. The AS5600 encoder draws 10 mA from the Uno R4 WiFi's onboard 3.3 V regulator, which is sourced from the 5 V LM2596 output; this load is negligible and within the Uno's onboard regulator rating.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.37	The Arduino Uno R4 WiFi shall be powered via the 5 V pin from the LM2596 buck converter output. The USB Type-B port shall remain available for firmware programming and serial debugging; during normal operation the USB cable may be unplugged.	Unlike the Mega2560 architecture, the Uno R4 WiFi can accept regulated 5 V on its 5V pin. Powering via the buck converter eliminates the laptop USB power dependency during standalone operation.	H		
HRS-3.1.38	The system shall have a dedicated power switch on the rear of the tower base that interrupts the 24 V supply from the barrel jack. A single switch cuts power to all components simultaneously.	A physical power switch is required for safe operation. Simultaneously cutting 24 V to both the TB6600 and the LM2596 input is safe; the Arduino enters an orderly power-off state without data corruption.	L		

3.1.7 Microcontroller and Firmware

The Arduino Uno R4 WiFi was selected as the embedded controller for the initial iteration of the Bland2Grand system. This board uses the Renesas RA4M1 microcontroller (Arm Cortex-M4, 48 MHz) and includes an integrated WiFi module (ESP-S3 based, 2.4 GHz 802.11 b/g/n). The onboard wireless capability simplifies the system architecture by eliminating the need for an external WiFi module and additional communication interfaces.

In this design iteration, the carousel-based dispensing mechanism requires two stepper motors. Both the NEMA 23 and NEMA 17 motors are controlled using TB6600 stepper motor drivers, which operate using a step/direction interface. Each driver requires two control pins (STEP and DIR) from the microcontroller. As a result, the two motors together require four digital I/O pins for motion control.

The Arduino Uno R4 WiFi provides 14 digital I/O pins and 6 analogue input pins, which appears sufficient for the motor drivers along with the additional sensors and control inputs expected in this version of the system. However, since this represents an early design iteration, the final pin allocation and controller selection may be revisited as the hardware

configuration and system requirements are further refined.

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.39	The embedded controller shall be an Arduino Uno R4 WiFi (Renesas RA4M1, Arm Cortex-M4, 48 MHz, 32 KB SRAM, 256 KB Flash; onboard ESP-S3 WiFi module, 2.4 GHz 802.11 b/g/n).	Onboard WiFi eliminates the external ESP8266 and its serial multiplexing complexity. The Cortex-M4 has sufficient processing speed for simultaneous encoder polling, HX711 reading, and WiFi communication without blocking.	H		
HRS-3.1.40	The firmware shall be written in C++ using the Arduino framework. Libraries required: AccelStepper (NEMA 17 auger motor control with acceleration profiles); AS5600 (I2C encoder readout); HX711 bogde (load cell reading); WiFiS3 (onboard WiFi communication for Uno R4 WiFi); ArduinoJson (JSON command parsing).	All libraries are open source with active maintenance. WiFiS3 is the official library for the Uno R4 WiFi's onboard module. AS5600 library (RobTillaart) confirmed compatible with the RA4M1 I2C peripheral.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.41	The firmware shall implement a non-blocking main loop. Carousel step generation, NEMA 17 auger stepping, HX711 polling, encoder I2C reads, and WiFi communication shall all be handled by interrupt-driven or timer-based scheduling without any operation blocking the others.	A blocking loop using `delay()` would prevent real-time scale feedback during motor operation. Non-blocking architecture using `AccelStepper.run()` and polled HX711 sampling at 10 Hz is sufficient.	H		
HRS-3.1.42	The firmware shall expose a JSON command interface over WiFi. Flask sends `{"carousel": N, "grams": X.X}`; the firmware responds with `{"status": "done", "actual": Y.Y}` upon completion.	JSON chosen because it is self-describing, extensible, and easily parsed on both the Arduino (ArduinoJson) and Flask (Python `json` module) sides. The `carousel` field specifies which module to index (1–8).	H		
HRS-3.1.43	The firmware shall implement a carousel sequencing algorithm that, for a multi-spice blend, sorts the required modules by index angle and plans the shortest-path rotation sequence to minimise total carousel travel.	For a 5-spice blend, naive sequential indexing could require up to 5 full or partial carousel rotations. Sorting by angular order and rotating the carousel in one direction reduces total travel, saving time.	M		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.44	The firmware shall implement an emergency stop that halts the carousel and engagement motors and closes the WiFi connection if no command is received from Flask within a 30-second timeout window during an active dispense session.	Prevents indefinite motor operation if the laptop loses WiFi connectivity or Flask crashes mid-dispense.	M		
HRS-3.1.52	The engagement motor (NEMA 17) shall operate open-loop with respect to rotational position. No encoder feedback is used on the auger shaft. The HX711 load cell is the sole feedback mechanism controlling when the engagement motor stops. The firmware relies entirely on the measured weight reaching the target value to terminate each dispense, not on a step count or time estimate.	Open-loop stepper control is standard practice for auger dispensing when a scale provides closed-loop weight feedback. A rotary encoder on the auger shaft would add complexity and cost without benefit, since the load cell already provides the ground truth. Step loss on the engagement motor does not cause a safety hazard; it simply results in less spice dispensed, which the load cell detects and corrects by continuing to run the motor.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.53	The firmware shall implement a per-spice dispense timeout. If the target weight for a single spice has not been reached within 60 seconds of the engagement motor starting, the firmware shall stop the engagement motor, report a `{"status": "timeout", "actual": Y.Y}` response to Flask, and abort the remainder of the blend dispense.	Prevents the engagement motor from running indefinitely if a module is empty, jammed, or the load cell malfunctions. 60 seconds is chosen as a conservative upper bound: at the slowest calibrated dispense rate (oregano at low step rate), a 50 g dispense should complete in under 45 seconds. If 60 seconds elapses without reaching target, a fault condition is assumed. Flask shall display an appropriate user-facing error message.	H		

3.1.8 Recipe Intelligence and Software

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.45	The Flask backend shall maintain a local SQLite database containing a minimum of 100 pre-populated dish recipes, each specifying the dish name and the gram amount per serving of each of the 8 supported spices.	SQLite selected over a full relational database server because the dataset is small, the application is single-user, and SQLite requires zero server configuration.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.46	When a user submits a dish name, the Flask backend shall first query the local SQLite database for a case-insensitive partial match. If a match is found, the result shall be returned with no external network request, targeting a response time under 100 ms.	Local database lookup avoids AI API latency (typically 1–3 seconds) for common dishes.	H		
HRS-3.1.47	If no match is found in the local database, the Flask backend shall issue a request to an AI language model API with a structured prompt requesting a JSON-formatted spice blend for the submitted dish name. The returned blend shall be validated against the 8 available spice channels and saved to the local database before being returned to the user.	The save-on-first-use behaviour means the AI API is called at most once per unique dish name. All subsequent requests for the same dish are served from the local database instantly.	M		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.1.48	The Flask backend shall expose the following HTTP endpoints: GET /search returning up to 3 matching blends; POST /dispense accepting a blend ID, serving count, and optional module sequence optimisation flag; GET /status returning current dispense progress as a JSON object; POST /calibrate accepting motor index and grams-per-revolution for calibration updates; POST /recipe accepting custom recipe definitions from the user.	RESTful HTTP API using Flask routing decorators. All endpoints return JSON. The dispense endpoint passes carousel module indices (1–8) and gram targets to the firmware.	H		
HRS-3.1.49	The Flask backend shall be accessible from any device on the same WiFi network by navigating to `http://{laptop_ip}:5000` in a standard mobile browser. No app installation shall be required on the user's phone.	Hard constraint from the user characteristics profile. Any installation step would be a barrier to adoption for User Profile A.	H		

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Req#	**Requirement**	**Comments**	**Pri.**	**Date Rvwd**	**Customer Rev. / Apvd**
HRS-3.1.50	The web interface shall update the dispense progress screen in real time using server-sent events (SSE) or polling at 2 Hz, showing the current dispensed weight and target weight for the spice currently being dispensed, and indicating which carousel module is currently active.	SSE is the preferred push model. The progress screen shall also visually indicate when the carousel is rotating vs. when dispensing is active, so observers can follow the machine's state.	M		
HRS-3.1.51	The system shall support a custom recipe builder in the web interface allowing the user to define a blend name, specify gram amounts per serving for any combination of the 8 spices, and save the recipe to the local database.	Addresses User Profile B (commercial operator) who needs to store proprietary house recipes for repeatable large-batch use.	M		

3.1.9 Spice Module Configuration

The following 8 spices were selected after analysing spice frequency across 4 major world cuisines (Mexican, Italian, Indian/South Asian, and North American BBQ/Cajun). The selection criterion was to maximise the number of complete, commonly used spice blends achievable from a fixed set of 8 channels.

Slot	Spice	Primary Cuisines	Bulk Density (g/mL)	Blends Enabled
1	Cumin	Mexican, Indian, Middle Eastern	0.53	7
2	Paprika	Spanish, Hungarian, BBQ, Cajun	0.55	6
3	Garlic Powder	Universal	0.49	8
4	Chili Powder	Mexican, Tex-Mex, BBQ	0.50	5
5	Oregano	Italian, Greek, Mexican	0.19	4
6	Onion Powder	Universal	0.51	8
7	Black Pepper	Universal	0.43	8
8	Cayenne	Universal heat addition	0.45	6

Oregano has a significantly lower bulk density (0.19 g/mL). The auger will need to spin approximately 2.5 times longer to dispense the same gram amount of oregano as cumin. This is handled automatically by the closed-loop scale feedback system, but the per-spice calibration step for oregano requires a correspondingly lower target step rate to prevent excessive speed that could break up the fragile leaf structure.

3.2 User Interface Requirements

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.2.1	The web interface shall consist of a single-page application served by Flask. Navigation between screens (search, results, serving size, dispensing progress, completion) shall occur via JavaScript DOM manipulation without full page reloads.	Full page reloads between steps would interrupt the server-sent events connection used for real-time progress updates.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.2.2	The search screen shall present a single text input field and a Search button. As the user types, the interface shall display autocomplete suggestions from the local database at a debounce interval of 300 ms.	300 ms debounce is standard UX practice. Below 300 ms requests fire too rapidly on mobile keyboards; above 500 ms the interface feels sluggish.	H		
HRS-3.2.3	The results screen shall present a maximum of 3 blend options for the searched dish, each displayed as a card showing the blend name, a list of spices and their amounts per serving, and a Select button.	Presenting more than 3 options introduces choice paralysis for User Profile A who is already seeking guidance.	H		
HRS-3.2.4	The serving size screen shall present a numeric slider from 1 to 20 servings. All gram amounts displayed on screen shall update in real time as the slider moves, calculated client-side without a server round-trip.	Client-side calculation reduces server load and makes the slider feel responsive. Formula: displayed grams = base grams/serving $\times$ serving count.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.2.5	The dispense progress screen shall display one progress bar per spice being dispensed, showing current dispensed weight versus target weight. The screen shall also display a carousel position indicator showing which of the 8 module slots is currently active. Completed spices shall show a checkmark and the active spice shall be visually highlighted.	The carousel position indicator differentiates this system's UI from a static multi-motor design, providing visible feedback that the machine is physically rotating between modules.	H		
HRS-3.2.6	The USB Type-B port on the Uno R4 WiFi shall remain accessible from outside the enclosure for firmware re-programming and serial debug output. The port shall not be covered or obstructed by the 3D-printed tower body.	Required for TEJ4M B5.4 (low-level programming on a microcontroller device).	H		

3.3 Hardware Interfaces

3.3.1 Internal Interfaces

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.3.1.1	The Uno R4 WiFi shall drive the TB6600 carousel driver via 2 digital output pins: STEP on pin 3 and DIR on pin 4. Signal level shall be 3.3 V logic (Uno R4 WiFi RA4M1 I/O is 3.3 V). The TB6600 input threshold is 3.5 V minimum; a pull-up resistor or signal conditioning shall be added if logic-low-to-high transitions are not detected reliably.	The RA4M1 I/O is 3.3 V. The TB6600 input low threshold is 0.8 V and high threshold is 3.5 V, placing the 3.3 V output level marginally below the high threshold. A 3.3 V-to-5 V logic level shifter on the STEP and DIR lines shall be included in the design as a mitigation.	H		
HRS-3.3.1.2	The Uno R4 WiFi shall drive the NEMA 17 TB6600 engagement driver via 2 digital output pins: STEP on pin 5 (PWM-capable) and DIR on pin 7. The TB6600 supply voltage shall be 24 V, consistent with the carousel TB6600.	The same 3.3 V-to-5 V logic level shifter used for the carousel TB6600 STEP and DIR lines shall also condition the NEMA 17 TB6600 STEP and DIR lines. No separate level shifter is required.	H		
HRS-3.3.1.3	The AS5600 encoder shall communicate with the Uno R4 WiFi via hardware I2C on pins A4 (SDA) and A5 (SCL) at 400 kHz. Both SDA and SCL shall have 4.7 kΩ pull-up resistors to 3.3 V.	The AS5600 operates at 3.3 V and the Uno R4 WiFi I2C bus is 3.3 V, so no level shifting is required on this interface.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.3.1.4	The HX711 shall communicate with the Uno R4 WiFi via bit-banged 2-wire protocol on pins 9 (DT) and 10 (SCK). The HX711 shall be configured for 10 SPS by tying the RATE pin to GND.	10 SPS provides one new reading per 100 ms control cycle. The 80 SPS mode increases noise by ≈3× and is not needed.	H		
HRS-3.3.1.5	The load cell shall be connected to the HX711 board via a 4-wire colour-coded cable (red: E+, black: E−, white: A−, green: A+). Wire length between the load cell and HX711 shall not exceed 200 mm.	The full-bridge output at full scale is ≈1.0 mV/V × 5 V excitation = 5 mV. Cables beyond 200 mm begin to act as antennas picking up switching noise from the stepper motor drivers.	H		
HRS-3.3.1.6	All internal signal wiring between the Uno R4 WiFi, motor drivers, HX711, and AS5600 shall use 22 AWG solid-core jumper wire. All wiring on the 24 V power rail and motor power leads shall use 18 AWG stranded wire.	Voltage drop for 18 AWG at 2.0 A over 300 mm: $V = IR = 2.0 \times 0.00627 = 0.013$ V at 24 V supply. Negligible. 18 AWG provides mechanical robustness for higher-current power runs.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.3.1.7	A 3.3 V-to-5 V bidirectional logic level shifter module shall be installed between the Uno R4 WiFi and both TB6600 drivers. The level shifter shall condition all 4 signal lines: STEP (pin 3) and DIR (pin 4) for the carousel TB6600, and STEP (pin 5) and DIR (pin 7) for the engagement TB6600. A single 4-channel level shifter module (e.g., TXS0104E or BSS138-based) is sufficient to handle all 4 lines simultaneously.	The RA4M1 outputs 3.3 V logic. The TB6600 requires a minimum 3.5 V high-level input to reliably detect a logic-high. Without level shifting, the 0.2 V margin is insufficient and may cause missed steps or erratic motor behaviour, particularly at higher step frequencies. The level shifter guarantees 5 V signals at both TB6600 inputs, eliminating this marginal condition entirely. This is a required component, not an optional mitigation.	H		

3.3.2 External Interfaces

Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.3.2.1	The system shall accept AC mains power (100–240 V, 50/60 Hz) via the 24 V 6 A AC/DC supply. The supply output barrel jack (5.5 × 2.1 mm) shall connect to a barrel jack screw-terminal adapter feeding the 24 V internal distribution rail.	Universal input range ensures compatibility with both North American (120 V 60 Hz) and international (230 V 50 Hz) mains without modification.	H		
HRS-3.3.2.2	The Uno R4 WiFi shall provide a USB Type-B connector accessible on the exterior of the tower enclosure. This port shall be used for firmware programming, serial debug output, and optional laptop-powered operation during development.	The RA4M1 has a dedicated USB-CDC interface. During normal operation the USB cable may be disconnected and the board powered from the 5 V LM2596 output.	H		
HRS-3.3.2.3	The system shall communicate with the host laptop Flask server over a 2.4 GHz 802.11 b/g/n WiFi network using the Uno R4 WiFi's onboard WiFi module in station mode.	Station mode allows the phone, laptop, and Arduino to all communicate via the existing network infrastructure without requiring the phone to associate directly with the Arduino, which would disconnect the phone from the internet and prevent AI API calls.	H		

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Req#	Requirement	Comments	Pri.	Date Rvwd	Customer Rev. / Apvd
HRS-3.3.2.4	The user shall interact with the system via a standard mobile web browser at `http://{laptop_ip}:5000`. No app installation, Bluetooth pairing, or cable connection is required from the user device.	Zero-installation requirement is fundamental to User Profile A. Any installation step would be a barrier to adoption for a beginner user.	H		
HRS-3.3.2.5	The load cell scale platform shall be an external-facing surface of the tower base, sized to accommodate a bowl with a base diameter of up to 150 mm. The platform shall be level to within 2 degrees to ensure accurate load cell readings.	At 2 °, cosine error = $1 - \cos(2) = 0.06\%$, corresponding to 0.06 g error on a 100 g measurement, well within $\pm$0.3 g. At 5 °, error grows to 0.38%, approaching the accuracy limit.	H		
HRS-3.3.2.6	Each spice module shall present an externally accessible twist-off lid at its top face, allowing the user to refill the spice chamber without removing the module from the carousel. The lid shall provide a positive retention mechanism to prevent accidental opening during carousel rotation.	Ease of refilling is a key usability requirement. The lid retention mechanism must resist the centripetal acceleration of carousel rotation at maximum indexing speed ($\approx$0.5 g radial at 60 RPM at 100 mm radius).	M		

4 Requirements Confirmation / Stakeholder Sign-Off

Meeting Date	Attendees (Name and Role)	Comments
March 8, 2026	Designer (Student), Manager (Teacher)	Initial HRS v1.0 review. Requirements HRS-3.1.1 through HRS-3.3.2.6 reviewed. Version 1.0 approved.
April 7, 2026	Designer (Student), Manager (Teacher)	Phase 1 checkpoint. Carousel motor, encoder, and engagement motor breadboard confirmed functional. Load cell integrated.
April 28, 2026	Designer (Student), Manager (Teacher)	Phase 2 checkpoint. Flask-to-Arduino WiFi communication confirmed. Recipe search and carousel command interface functional.
May 19, 2026	Designer (Student), Manager (Teacher)	Phase 4 checkpoint. Full system integration complete. Carousel indexing and closed-loop dispense verified for all 8 modules.
June 7, 2026	Designer (Student), Manager (Teacher), Peers	Final demonstration and submission. All HRS requirements confirmed satisfied.

A Definitions, Acronyms, and Abbreviations

Term	Definition
ADC	Analog-to-Digital Converter. A circuit that converts a continuous analog voltage signal to a discrete digital numerical value.
AS5600	A 12-bit contactless magnetic rotary position sensor (encoder) from AMS, communicating over I2C. Used in this project to determine the absolute angle of the carousel shaft.

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Term	Definition
Auger	A helical screw mechanism used to convey powdered spice along a tube by rotation. See Appendix C for a visual reference.
Carousel	The motorised rotating octagonal platform that holds the 8 spice modules and indexes each module to the dispensing position.
Flask	A lightweight Python web framework used to build the Bland2Grand backend server and serve the mobile web interface.
FDM	Fused Deposition Modelling. The 3D printing process used for all mechanical components in this project.
HRS	Hardware Requirements Specification. This document.
HX711	A 24-bit ADC integrated circuit designed specifically for reading Wheatstone bridge load cells.
I2C	Inter-Integrated Circuit. A 2-wire serial communication bus protocol (SDA + SCL) used in this project for the AS5600 encoder.
LM2596	A DC-DC buck (step-down) converter IC rated to 3 A output. Used in this project to step 24 V down to 5 V for logic components (Arduino Uno R4 WiFi and HX711).
NEMA 17	A stepper motor frame size standard. In this project, the NEMA 17 (17HS4401S) is the engagement motor that drives each module's auger shaft.
NEMA 23	A larger stepper motor frame size. In this project, the NEMA 23 (23HS5628) is the carousel indexing motor.
PLA	Polylactic Acid. A biodegradable thermoplastic polymer used as the 3D printing filament. Heat deflection temperature approximately 60 ℃.
RA4M1	The Renesas ARM Cortex-M4 microcontroller at 48 MHz at the heart of the Arduino Uno R4 WiFi.
SQLite	A lightweight, serverless relational database engine. Used for the Flask recipe database.
SSE	Server-Sent Events. An HTTP protocol allowing a server to push real-time updates to a browser client.
Stepper Motor	A brushless DC motor that divides a full rotation into a fixed number of discrete steps, allowing precise angular positioning without a position feedback sensor.

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Term	Definition
Tare	The process of zeroing a scale to exclude the container weight. Performed before each individual spice dispense.
TB6600	A bipolar stepper motor driver module rated at up to 4 A and 42 V. Used in this project to drive both the NEMA 23 carousel motor and the NEMA 17 engagement motor.
WiFi	Wireless Fidelity. The 802.11 wireless LAN standard. The Uno R4 WiFi's onboard module supports 802.11 b/g/n on 2.4 GHz.

B References

#	Reference
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2	Arduino Uno R4 WiFi Documentation. Arduino. Renesas RA4M1, 48 MHz, 32 KB SRAM, 256 KB Flash, 14 digital I/O, onboard 2.4 GHz WiFi. Available at: Uno R4 WiFi
3	NEMA 23 (23HS5628) Stepper Motor Datasheet. 2.8 A, 2-phase, 1.8° step angle, holding torque 126 N·cm, 8 mm output shaft. Available at: 23HS5628 Datasheet
4	NEMA 17 (17HS4401S) Stepper Motor Datasheet. 1.5 A, 4-lead, 1.8° step angle, holding torque 45 N·cm, 5 mm output shaft. Available at: 17HS4401S Datasheet
5	TB6600 Stepper Motor Driver Datasheet. Toshiba TB6600HG. Input voltage 9–42 V, output current up to 4 A, 8 microstepping modes. Available at: TB6600HG Datasheet
6	AS5600 Magnetic Rotary Position Sensor Datasheet. AMS-OSRAM. 12-bit resolution, I2C interface, 3.3–5 V supply, programmable zero position. Available at: AS5600 Datasheet
7	HX711 24-bit ADC Datasheet. Avia Semiconductor. Rev 1.0. 24-bit resolution, 10/80 SPS selectable, 2.6–5.5 V operating voltage. Available at: HX711 Datasheet
8	NOYITO 1 kg Load Cell Specification. Output sensitivity 1.0 mV/V, rated load 1000 g, safe overload 150% rated load, operating temperature −20 to +65 °C. Available at: NOYITO 1kg Load Cell

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#	**Reference**
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10	AccelStepper Library Documentation. Mike McCauley, AirSpayce Pty Ltd. Version 1.64. Available at: airspayce.com/mikem/arduino/AccelStepper
11	RobTillaart AS5600 Arduino Library. Available at: github.com/RobTillaart/AS5600. I2C driver confirmed compatible with RA4M1.
12	bogde HX711 Arduino Library. Available at: github.com/bogde/HX711. Confirmed compatible with Uno R4 WiFi.
13	WiFiS3 Library Documentation. Arduino. Official WiFi library for Arduino Uno R4 WiFi onboard module. Available at: docs.arduino.cc/libraries/wifis3
14	Flask Web Framework Documentation. Pallets Projects. Version 3.x. Available at: flask.palletsprojects.com

C Auger Reference Diagram

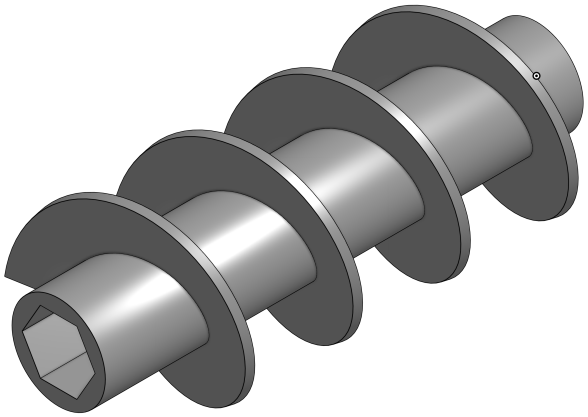

Figure 2: Example auger screw mechanism showing helical flights and central shaft.